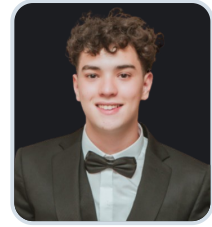


Sofiène Zayati

Full-Stack & AI Engineer | Embedded Systems Background

Tunisia · Open to relocation to Germany



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PROFESSIONAL PROFILE

Full-stack and AI engineer pursuing an Engineering Degree in Web and Internet Technologies at ESPRIT. I build complete products, document-intelligence workflows, automation, Python and Node.js services, and connected C++ devices using MQTT, BLE, and ESP32/WT32. My work spans interfaces, APIs, data, decision logic, deployment, and hardware integration. Goethe-Zertifikat A2; targeting software, AI automation, or IoT roles in Germany.

PROFESSIONAL EXPERIENCE

Jul 2025 – Aug 2025

AI Automation & Full-Stack Developer

Inspark Connect · Tunis, Tunisia · Professional placement

- Built n8n and Gemini assistants integrated into an existing Laravel, Vue.js, and MySQL e-commerce platform.
- Designed intent routing, JSON validation, SQL aggregation, product enrichment, campaign, sales, and logistics workflows.

Feb 2024 – May 2024

Embedded Systems & Software Developer

Scheidt & Bachmann Maghreb · Mégrine, Tunisia · Graduation internship

- Engineered and validated MacroPark's admin platform, Flutter access, LPR/MQTT integration, and WT32 barrier control across five Scrum sprints.
- Implemented and tested BLE/MQTT access paths, persistent controller configuration, OTA maintenance, and Docker Compose deployment.

Jan 2023 – Feb 2023

Hardware & Software Project Intern

Scheidt & Bachmann Maghreb · Mégrine, Tunisia

- Designed and prototyped a connected smart-lock system for the company.

Jan 2022 – Feb 2022

Web Development Intern

Sotunol · Mégrine, Tunisia

- Developed and improved the company's web platform.

EDUCATION

Sep 2024 – Present

Engineering Degree in Web and Internet Technologies (TWIN)

ESPRIT School of Engineering · Ariana, Tunisia

Sep 2021 – Jun 2024

Bachelor's Degree in Embedded and Mobile Systems

Higher Institute of Technological Studies of Radès · Radès, Tunisia

Sep 2017 – Jun 2021

Technical Sciences Baccalaureate

Lycée Technique Farhat Hached · Radès, Tunisia

SELECTED TECHNICAL PROJECTS

Machetamache — Local-First Recipe & Food-Costing Product Product designer & full-stack engineer · Personal product for a working chef - Designed and built a React 19, TypeScript, Express 5, and SQLite product for recipes, exact food costing, portion scaling, printing, backups, and PC/phone LAN access. - Added integer-based currency calculations, Vitest/Supertest coverage, network-safety controls, legacy-phone fallbacks, and a portable Windows release. Case study: sofienezayati.tn/project/machetamache	2026
SmartProperty — AI-Powered Real Estate Platform Sole full-stack & AI engineer · End-to-end product ownership - Independently designed and built the complete multi-role product for agents, administrators, clients, and accountants across listings, discovery, visits, applications, payments, analytics, and operations. - Engineered React, NestJS, FastAPI, MongoDB, Socket.IO, XGBoost valuation, recommendation, CLIP/BLIP image intelligence, OCR solvency analysis, Groq assistance, and containerized delivery. Case study: sofienezayati.tn/project/smartproperty	2026
Prigado AI — Intelligent E-Commerce Automation AI automation & full-stack developer · Inspark Connect placement - Built conversational workflows for product creation, enrichment, marketing, sales reporting, and logistics analysis. - Used n8n, Gemini, Laravel, Vue.js, MySQL, SMTP, REST APIs, guarded JSON outputs, and structured SQL aggregation. Case study: sofienezayati.tn/project/prigado	2025
MacroPark — Connected Parking Access & Supervision Embedded systems & software engineer · Graduation project at Scheidt & Bachmann Maghreb - Delivered a complete admin, Flutter mobile, Quercus LPR, FastAPI/MQTT, and WT32-ETH01 parking system across five Scrum sprints. - Validated camera/MQTT and phone/BLE entry paths, four barrier channels, SPIFFS persistence, OTA updates, and physical relay control. Case study: sofienezayati.tn/project/macropark	2024

Additional documented work includes Green Coffee OS, Genius PMO HRMS, TounsiAds, Interna, InnoMall, Smart Agri, and embedded IoT systems. Explore the full portfolio at sofienezayati.tn.

TECHNICAL SKILLS

Programming	TypeScript, JavaScript, Python, Java, C, C++, SQL
Frontend	React, Angular, Vue.js, JavaFX, HTML, CSS, responsive interfaces
Backend & Automation	Node.js, NestJS, Express, FastAPI, Spring Boot, Symfony, Laravel, n8n, REST APIs, WebSockets
Data & Applied AI	MySQL, MongoDB, SQLite, Firebase, XGBoost, CLIP/BLIP, OCR, Gemini, Groq, LLM integration, explainable scoring
Embedded & IoT	ESP32, WT32-ETH01, MQTT, BLE, Ethernet/Wi-Fi, OTA updates, relay and sensor integration
DevOps & Tools	Git, GitHub, Docker, Docker Compose, Kubernetes, Jenkins, CI/CD, Linux

LANGUAGES

Arabic Native	English Fluent · self-assessed	French Fluent · self-assessed	German A2 · Goethe-Zertifikat
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